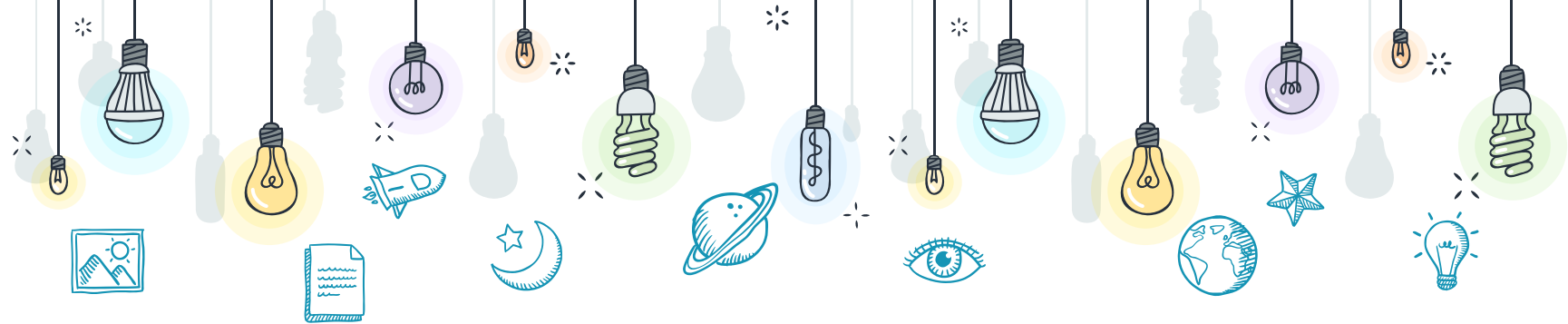




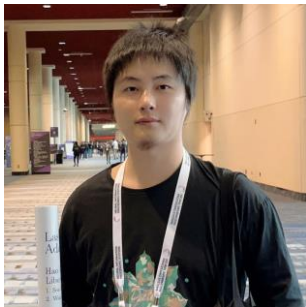
Evaluations and Benchmarks in Context of Multimodal LLM



<https://mllm2024.github.io/CVPR2025/>




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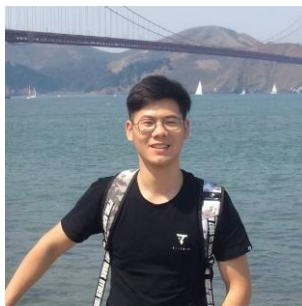
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* Part-IV

Video Capability Evaluation



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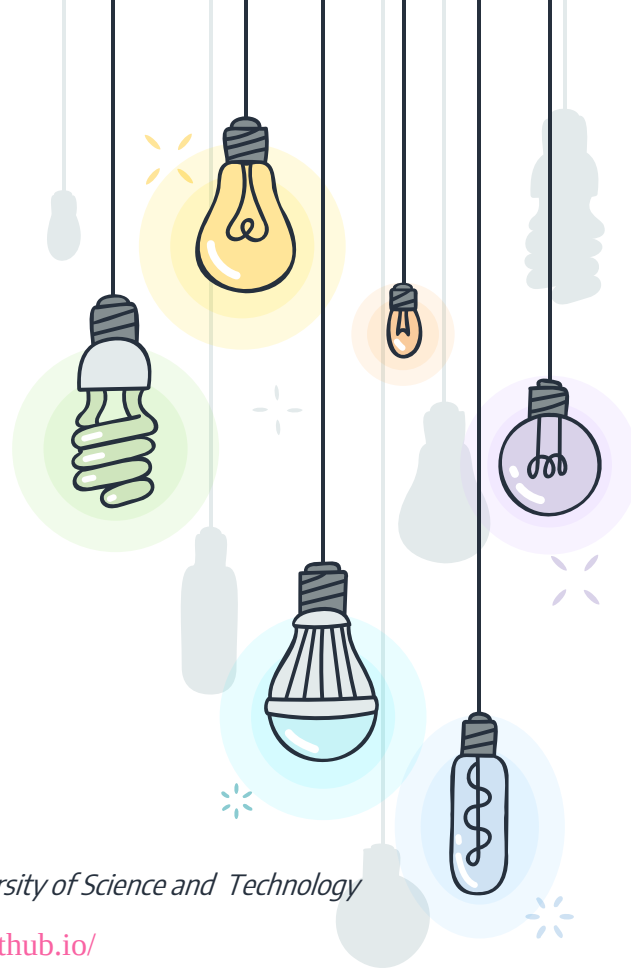
<https://zjuchenlong.github.io/>

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* Table of Content

+ Preliminary

- × Image and Video
- × Traditional Video Tasks
- × Transition to Video LLM

+ Multimodal Benchmarks of VideoLLM

- × Understanding
- × Reasoning
- × Generation
- × Application

+ Summary

* Image and Video

- ✦ Understanding image and video data are two of the most important things to help us navigate the physical world. They have strong connections in visual information but also a lot of distinctions in many aspects.



Image Understanding

- ✕ Static visual information
- ✕ Objects, attributes, spatial, etc.
- ✕ Natively unimodal and lower complexity

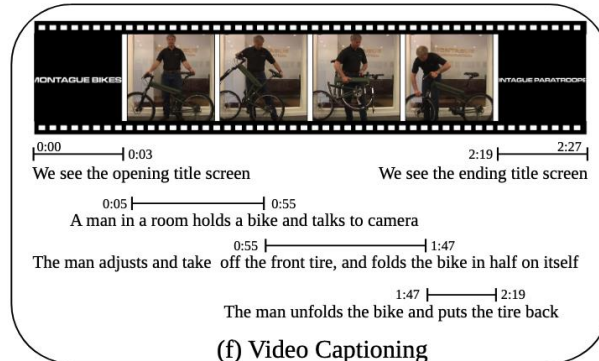
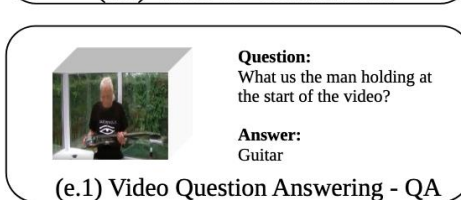
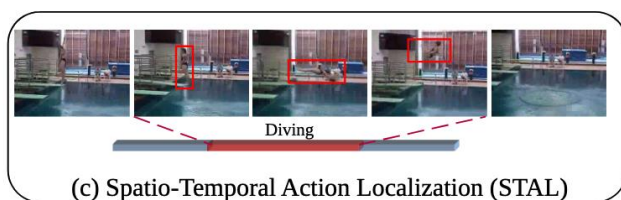
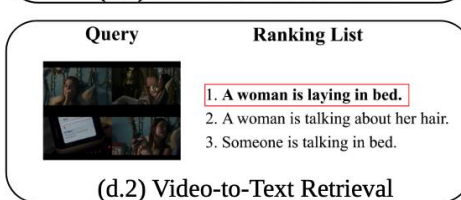
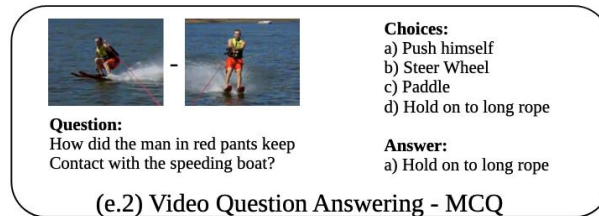


Video Understanding

- ✕ Dynamic visual information
- ✕ Events, trajectories, temporal, etc.
- ✕ Natively multimodal and higher complexity

* Traditional Video Understanding Tasks

- ✦ In the past decade, a large number of important video understanding tasks have emerged as separately studied lines of research, including video action recognition, temporal action localization, etc.



* Video Capability Benchmarks

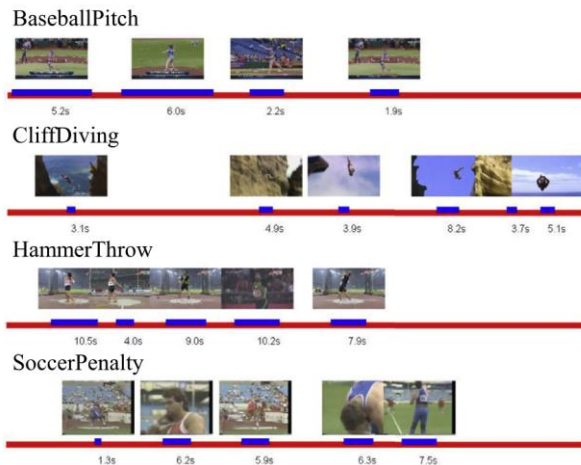
- ✦ A diverse range of downstream benchmarks are established to evaluate different aspects of video understanding capabilities, e.g., global content perception, local activity localization, visual scene recognition, and so on. However, it remains difficult to acquire and evaluate these capabilities **at the same time within a single model**.



Video Classification

- ✗ Global understanding
- ✗ Coarse-grained perception
- ✗

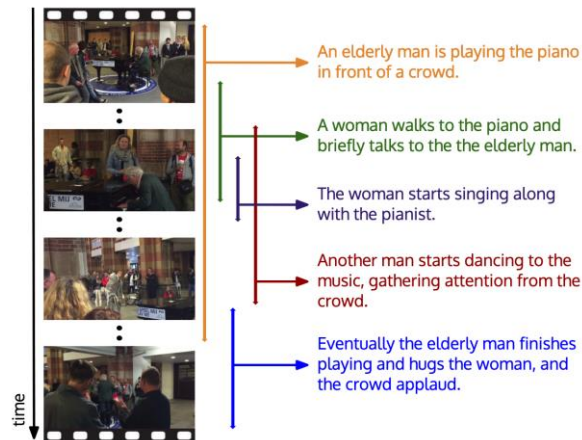
UCF-101, Kinetics-400 ...



Action Detection

- ✗ Local understanding
- ✗ Fine-grained perception
- ✗

THUMOS-14, ActivityNet v1.3 ...



Video Captioning

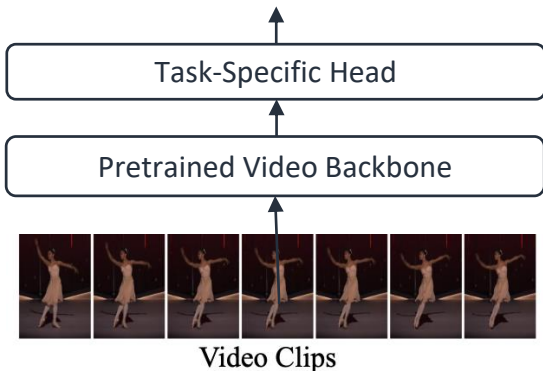
- ✗ Activity Understanding
- ✗ Scene recognition
- ✗

ActivityNet Captions, YouCook2, ...

* Specialist Video Models and VideoLLMs

- Large Language Models (LLMs) have advanced rapidly in recent years. Meanwhile, single-task specialist video models are being supplanted by Video Large Language Models (VideoLLMs). These VideoLLMs fuse LLMs' strong text-processing power into one interface that can handle many kinds of video tasks.

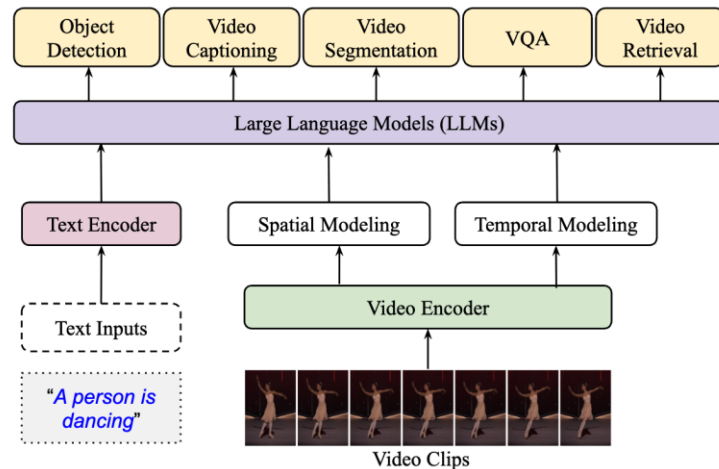
Single Task Output (e.g., action class labels)



Specialist Video Models

- ✗ Single-task capabilities
- ✗ Unimodal understanding
- ✗ Fixed output format

C3D, SlowFast, ...



Video Large Language Models

- ✗ Multi-task capabilities
- ✗ Multimodal understanding
- ✗ Flexible instruction following

Video-LLaVA, Video-ChatGPT, ...

* Multimodal Benchmarks of VideoLLM

+ Capability

- × Understanding
- × Reasoning
- × Generation
- × Application

* Multimodal Benchmarks of VideoLLM

+ Capability

- × Understanding
- × Reasoning
- × Generation
- × Application

Understanding

- + It refers to the ability to extract and integrate features from multimodal data to perform cross-modal analysis. This involves tasks such as interpreting visual representations, identifying key details, grasping semantic meaning, and responding accurately to related questions.
- × Visual Perception
- × Contextual Comprehension

* Understanding

Visual Perception

- + Visual perception capability is a foundational aspect of understanding benchmarks. It involves the ability to extract salient features and accurately recognize and interpret visual elements, such as multiple objects, text information, and complex emotional or implicit cues.
 - × Fine-grained Perception
 - × Coarse-grained Perception
 - × Low-level Perception

* Understanding

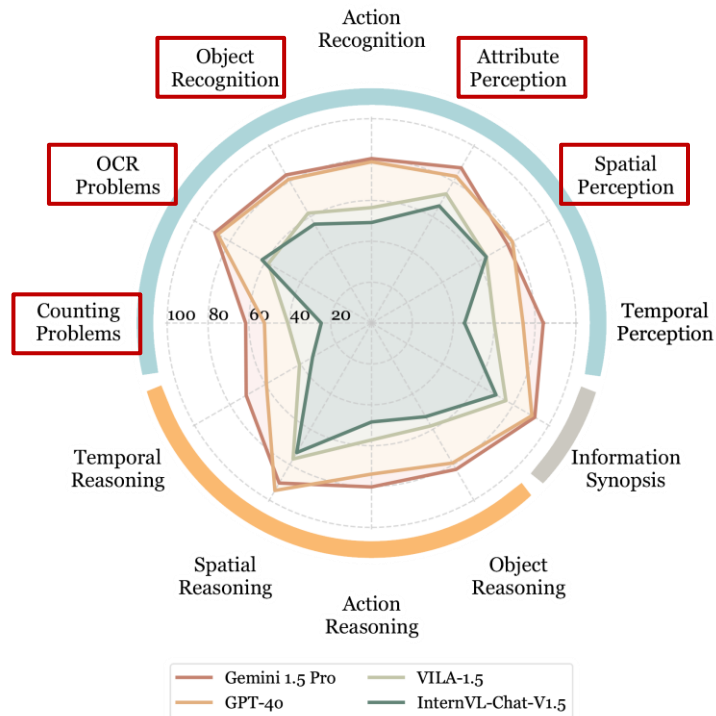
Fine-grained Perception



Q: What color are the nails of a woman with white hair?
A: Pink or purple.

- ✗ OCR
- ✗ Object recognition
- ✗ Attribute recognition
- ✗ Position recognition
- ✗ Counting
- ✗

Video-MME, Perception Test, MVBench, MMBench-Video,



Fine-grained Question Types in Video-MME

* Understanding

Coarse-grained Perception

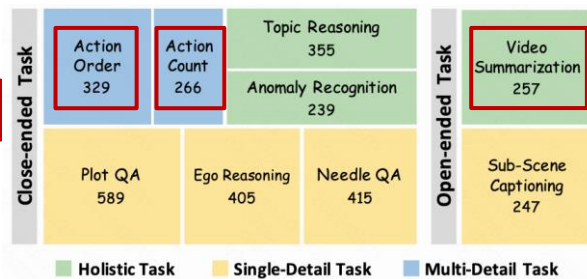
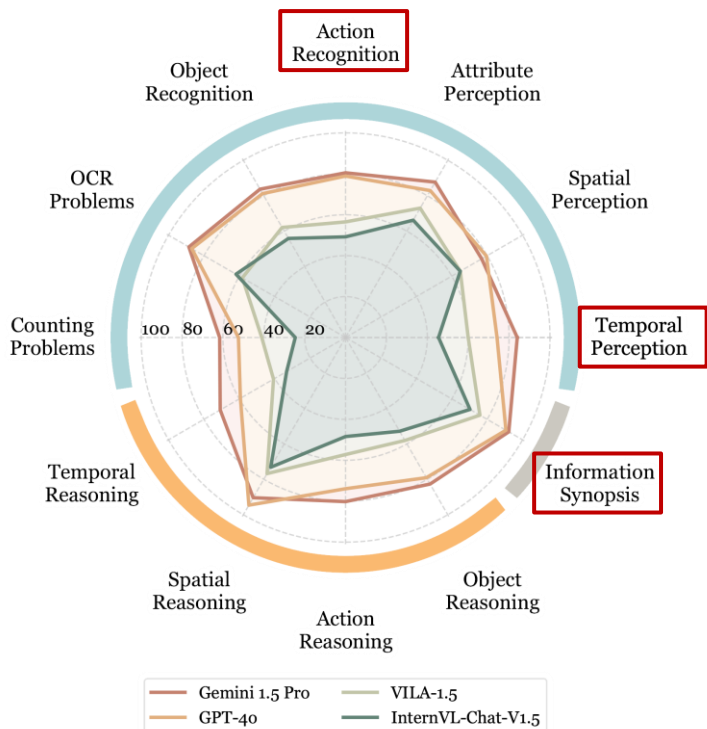


Q: What is this video about?

A: Tesla iOS App Integrates Shortcuts for Limitless Control

- × Scene
- × Event
- × Topic
- × Style
- × Emotion
- ×

MMBench-Video, Video-MME



* Understanding

Low-level Perception

Yes-or-No



Q: Does the fabric in this video exhibit good clarity and proper lighting?

A. Yes ✓ B. No

What-How



Q: What quality issues do not exist in this video?

A. None of the options B. Blurriness
C. Overexposure ✓ D. Underexposure

Open-ended



Q: Why is it difficult for viewers to identify the dish the man is cooking in the video?

Open-ended Response: The video has severe compression blur and block artifacts, significantly reducing the discernibility of objects in the video.

- × low-level degradation
- × Aesthetic distortions
- × Temporal Distortions
- ×

Q-Bench-Video, TempCompass, KVQ

Speed

Relative Speed



Meta Information

Subject: the yellow car
Speed: faster than other cars

Absolute Speed



Meta Information

Subject: entire video
Speed: slow motion

Direction

Object Direction



Meta Information

Subject: ball
Direction: rolling from left to right

Camera Direction



Meta Information

Subject: entire video
Direction: moving forward

Attribute Change

Size & Shape Change



Meta Information

Subject: ice cream
Attribute Change: melting

Color & Light Change



Meta Information

Subject: entire video
Attribute Change: getting darker

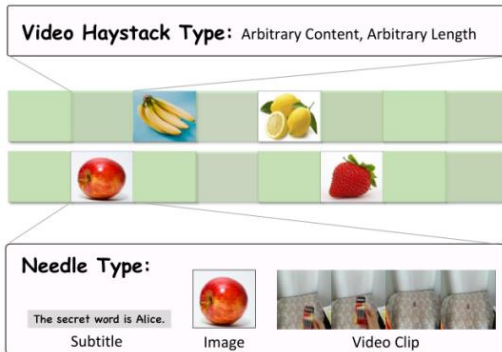
* Understanding

Contextual Comprehension

- + It refers to the ability of MLLMs to understand and interpret information that is influenced by the surrounding context. Based on the different input context formats, these benchmarks are grouped as follows:
 - × Long Context Understanding
 - × Interleaved Video-Language Understanding

* Understanding

Long Context Understanding

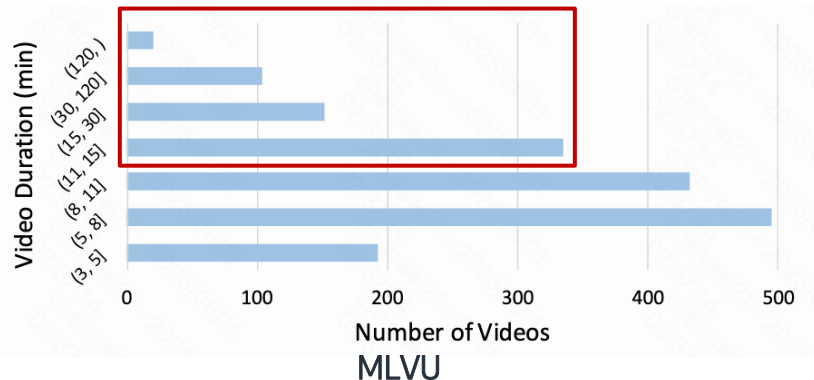
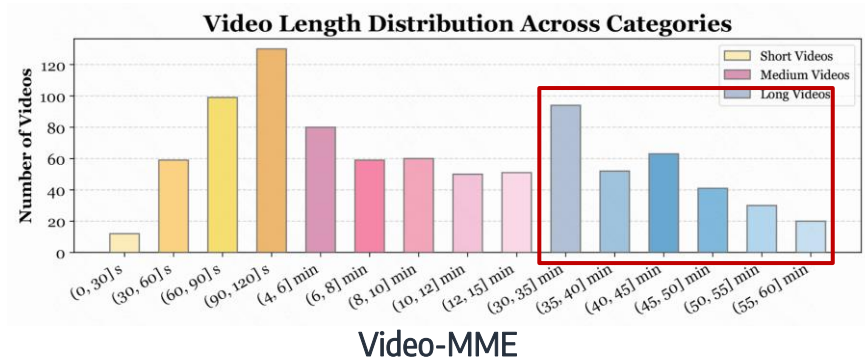


Question Type:

- Retrieval:** Asking for needle information
Required Ability: Temporal Perception
- Ordering:** Asking for the needle order
Required Ability: Chronological Order
- Counter:** Asking for the needle number
Required Ability: Temporal Coherence

- ✗ Needle In A Haystack
- ✗ Event Detection
- ✗

LongVideoBench, MLVU, Video-MME-M/L, VideoNIAH, EgoSchema, MMBench



* Understanding

Interleaved Video-Language Understanding



Question:

At the beginning of the video, a woman with a headband tied to her head, wearing a red top, carrying a black backpack, when the woman comes down from a hill with tall rocks, what changes occur to her backpack?

Options:

- A. There is a dark red jacket hanging on her black backpack
- B. Nothing changed
- C. There is a white jacket hanging on her black backpack
- D. There is a dark blue jacket hanging on her black backpack

- × Referred QA
- × Video-Text
- × Video-Audio
- × Video-Audio-Text
- × ...

LongVideoBench, Video-MME, VideoMMMU, MMVU...

Benchmark	Video frames	Subtitles / Text track	Raw audio
LongVideoBench	✓	✓ (all videos come with human subtitles)	✗ (audio not provided)
Video-MME	✓	✓ (subtitle text)	✓ (original audio track released)
Video-MMMU	✓	✓ (transcribed audio used as text)	✓ (lecture narration)
MMVU	✓	✓ ("Text + Video" category in dataset table)	✗ (no audio requirement stated)

* Understanding

Benchmark	Low-level	Fine-grained	Coarse-grained	Long-context	Interleaved V-L	Key focus
TempCompass	✓					Motion-tempo perception
Q-Bench-Video	✓					Video-quality distortions
VideoMME		✓	✓	✓	✓	Full-spectrum QA
MLVU			✓	✓	✓	Multi-task long-video
LongVideoBench				✓	✓	Hour-long interleaved QA
Video-MMMU				✓	✓	Lecture knowledge QA
MMVU				✓	✓	Expert-level reasoning

* Multimodal Benchmarks of VideoLLM

+ Capability

- × Understanding
- × Reasoning
- × Generation
- × Application

Reasoning

Understanding is necessary but not sufficient. Real-world queries demand causal, temporal, and commonsense inference.

* Reasoning

Reasoning: output intermediate thinking steps first, then predict the final answer.

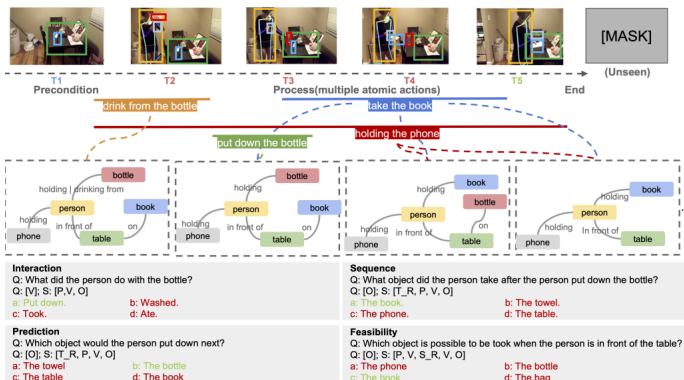
- + Many tasks in video require reasoning:
 - × **Long span:** evidence spread over hundreds of frames
 - × **Concurrent events:** many objects/actions at once
 - × **Occlusion:** key steps hidden or off-screen, must infer
 - × **Multi-step goals:** must summarise, plan, or predict next steps

* Reasoning

Reasoning Methods

Two types reasoning methods:

1. Base on Spatio-Temporal Scene Graph (STSG)



STEP (CVPR 25)
(2.5 + 1)D (AAAI 22)

2. End-to-end Reasoning (New trends)



Question: You are a robot beginning at the blue built-in bookshelf facing the blue built-in bookshelf. You want to navigate to the bathroom sink. You will perform the following actions (Note: for each [please fill in], choose either 'turn back,' 'turn left,' or 'turn right.'): 1. [please fill in] 2. Go forward until the door of the bathroom. 3. [please fill in] 4. Go forward until the sink. You have reached the final destination.
A. Turn Back, Turn Right B. Turn Left, Turn Left C. Turn Right, Turn Left D. Turn Back, Turn Left

Video-R1-7B: <think> Let me think about this carefully. The goal is to navigate from the blue built-in bookshelf to the bathroom sink. The first step is to determine the direction from the bookshelf to the bathroom.
1. The blue built-in bookshelf is on the left side of the room. The bathroom is on the right side of the room. So, I need to turn right from the bookshelf to head towards the bathroom.

2. After turning right, I need to go forward until I reach the door of the bathroom. This means I need to continue moving forward in the direction I just turned.

3. Once I reach the bathroom door, I need to turn left to enter the bathroom. This is because the sink is typically located on the left side of the bathroom.

So, the sequence of actions should be: 1. Turn right from the bookshelf. 2. Go forward until the bathroom door. 3. Turn left to enter the bathroom. 4. Go forward until the sink. (Perception and Reasoning)

Hmm, this seems to align with the sequence of actions in option C. Let's break it down one more time to verify.

Option A: Turn Back, Turn Right - This would mean going back to the bookshelf and then turning right, which doesn't make sense for reaching the bathroom.

Option B: Turn Left, Turn Left - This would mean turning left from the bookshelf and then turning left again, which also doesn't make sense.

Option D: Turn Back, Turn Left - This would mean going back to the bookshelf and then turning left, which is not the correct path. (Reflection)

Oh, I see. The correct sequence is indeed in option C. Let's confirm one last time. The correct sequence is to turn right, go forward to the bathroom door, turn left, and go forward to the sink. This matches option C. Final answer: C
Let's verify one last time. Yes, C is the correct answer. </think> <answer>C</answer>

Video-R1
VideoChat-R1
...

* Reasoning

Reasoning Methods

Based on evaluation mode:

Mode	What is fixed?	Typical metric	When it's used
Objective (automatic)	A canonical answer token (label, span, time-window, count...) for every question	Accuracy / F1, IoU, BLEU, etc.	Classical VQA-style benchmarks, long-horizon retrieval, physics & counting tasks
Subjective (human / LLM judgment)	A reasoning trace specification (e.g. step-wise CoT, key-frame evidence) but not an exact wording	Human vote or GPT-4 “faithfulness”/“helpfulness” score	Benchmarks that require models to show their work or justify answers

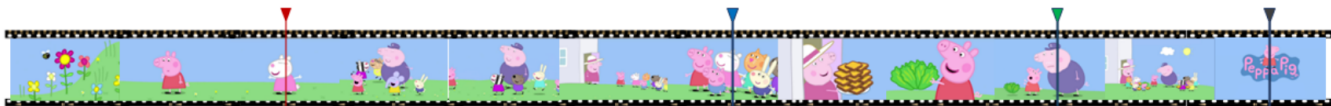
* Reasoning

Objective evaluation: closed questions solvable from explicit prompts or isolated visual cues.

Question: In which second does the character with the pink dress drop the item they are holding?

Options:

- A. Between the 90th and 92th seconds.
- B. Between the 70th and 72th seconds.
- C. Between the 20th and 22th seconds.
- D. Between the 50th and 52th seconds.



(from VideoVista)

Advantages

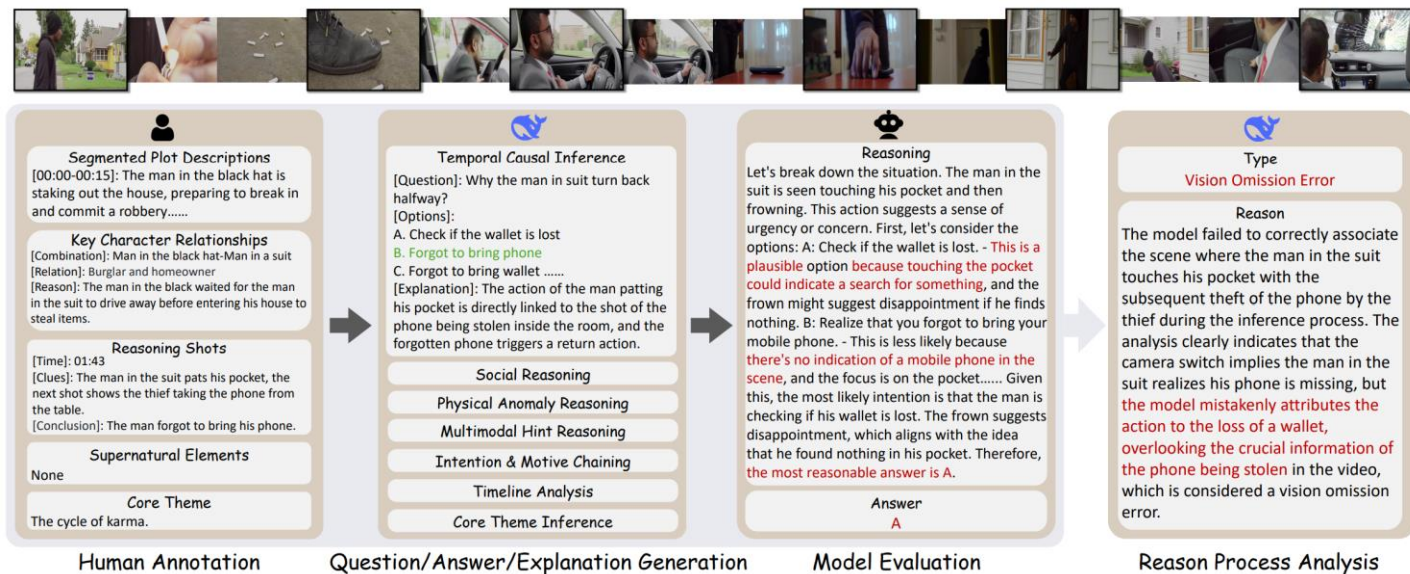
- Fully automated, fast scoring
- Single ground-truth label → easy comparison
- Accurately gauges basic perception & grounding
- Scales to very large datasets

Limitations

- Chain-of-thought ignored
- Guessing / dataset shortcuts still score high
- Weak on multi-step, long-range reasoning
- Little diagnostic insight when answers are wrong

* Reasoning

Subjective evaluation: each question comes with lots of comprehensive ground-truth reasoning trace; humans or an LLM judge how closely the model's predicted reasoning matches that trace and assign a subjective score.



(from Video-Holmes)

Objective vs. Subjective Video-Reasoning Evaluation

	Objective (answer-only)	Subjective (answer + reasoning trace)
What's judged	Single ground-truth label	Alignment between model's chain-of-thought and a gold reasoning trace
Strengths	• Fully automated → fast, cheap, large-scale • Clear, reproducible metrics	• Mirrors real human reasoning • Reveals how and why the model reaches its answer • Better for diagnosing hallucinations
Weaknesses	• Can reward lucky guesses / shortcuts • No insight into internal logic • Poor coverage of multi-step, long-range tasks	• High annotation and review cost • Slower, harder to standardise • Subjective scores may vary across raters / LLM assessors

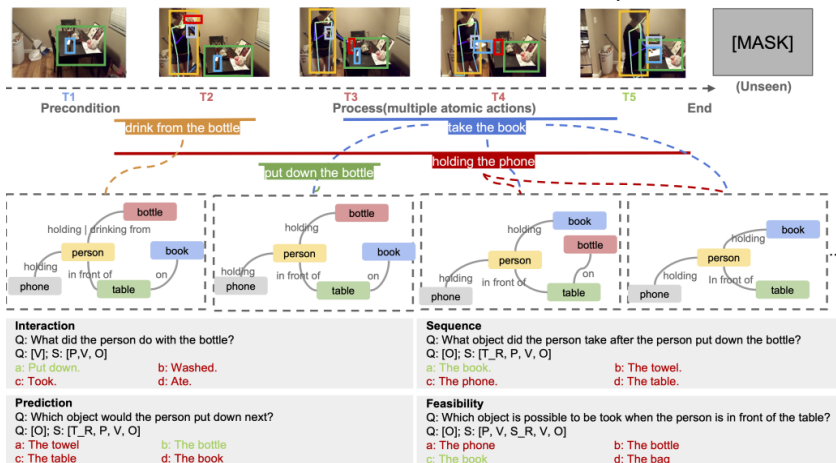
* Reasoning

- + Benchmarks can also be grouped by the target reasoning skill. It goes beyond basic understanding, encompassing the ability to perform complex inferences and draw logical conclusions across modalities. This includes tasks that require models to process and manipulate information, enabling them to solve problems and make decisions based on cross-modal data.
 - × Relational Reasoning
 - × Multi-step Reasoning
 - × Knowledge-based Reasoning

* Reasoning

Relational Reasoning

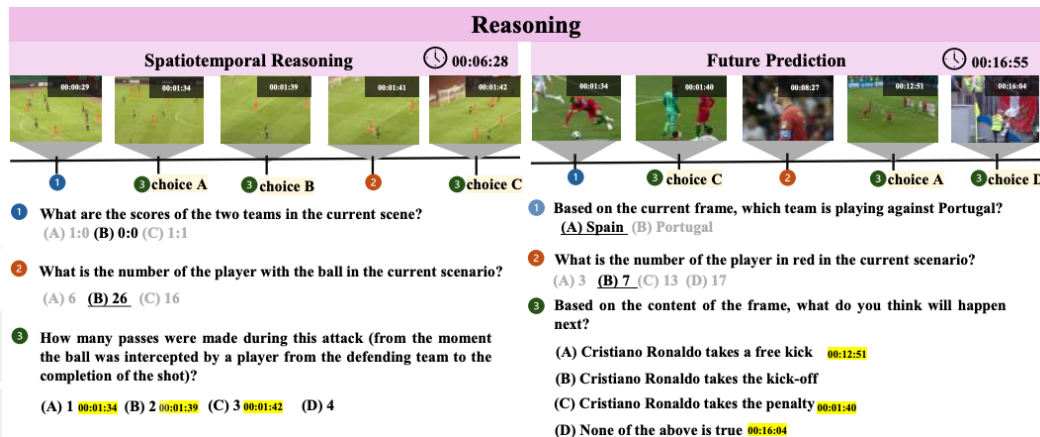
- ✦ Relational reasoning refers to the ability of MLLMs to recognize, manipulate, and reason about relationships between different entities or concepts.



Visual Spatial Relation

- ✦ Situated Reasoning
- ✦ Visual Relation Detection
- ✦

STAR, VideoVista ...



Temporal Relation

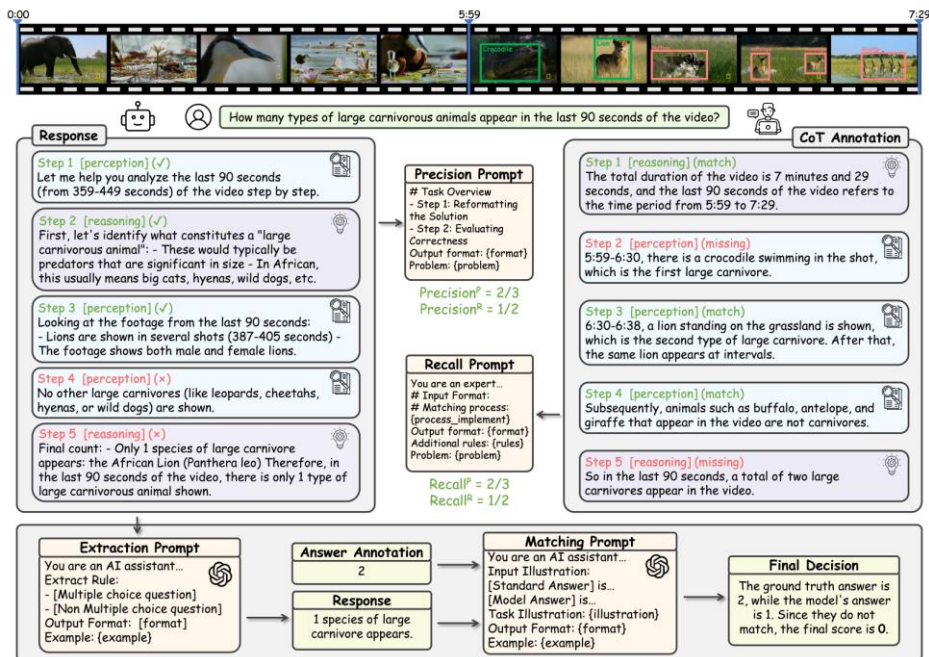
- ✦ Video QA
- ✦ Temporal grounding
- ✦

ViLMA, ReXTime, RTV-Bench ...

* Reasoning

Multi-Step Reasoning

- ✦ Multi-step Reasoning is crucial for complex cognitive tasks that necessitate navigating through a series of interconnected logical steps, e.g., chain of thoughts (CoT, which decomposes complex tasks into simpler, manageable subtasks).



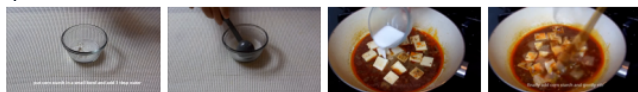
Chain-of-Thoughts

VideoCoT, VCR-Bench ...

* Reasoning

Knowledge-based Reasoning

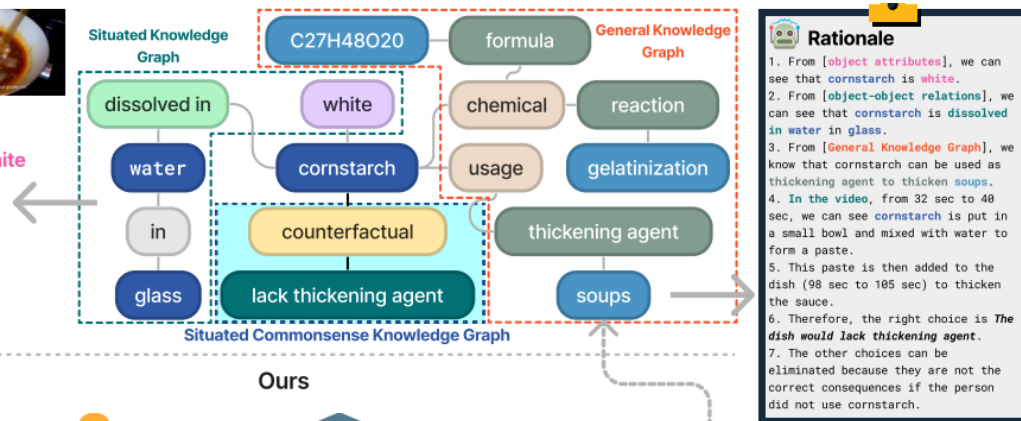
- ✦ Knowledge-based reasoning is the process of drawing conclusions or making decisions by leveraging structured information from an external knowledge graph, often incorporating commonsense knowledge answer complex queries.



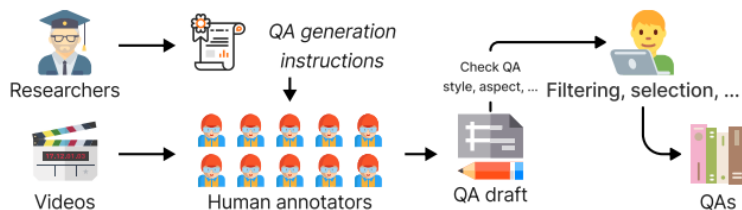
Mapo Tofu **attributes+relationship+counterfactual**

Question: What would happen if the person did not use the **white substance dissolved in water in glass**?

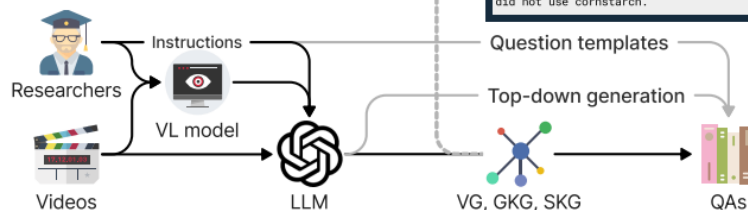
- A. The dish would be completely different.
- B. The dish would lack thickening agent. ✓
- C. The dish would lack seasoning and taste bland.
- D. The dish would lose its signature flavor.



Previous methods



Ours



Knowledge-based Reasoning

SOK-Bench ...

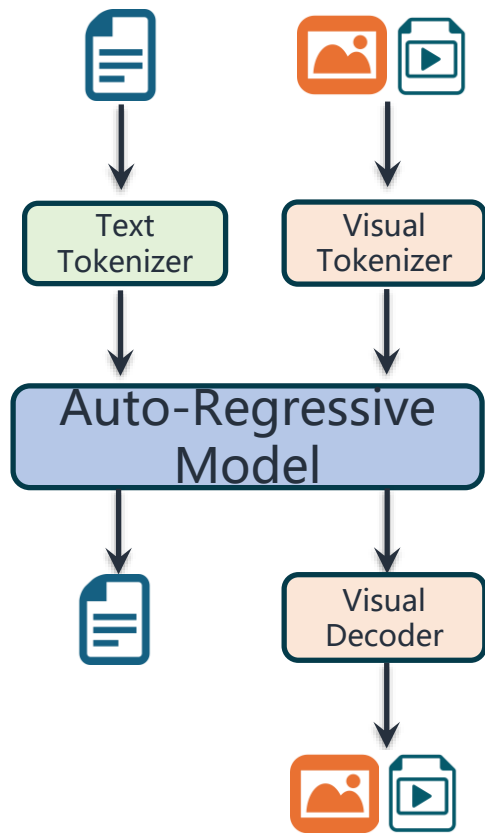
* Multimodal Benchmarks of VideoLLM

+ Capability

- × Understanding
- × Reasoning
- × **Generation**
- × Application

* Generation

+ Current MLLM Frameworks that Supports Video Generation



Discrete Tokenizer (e.g., VQ-VAE)

- VILA-U: a Unified Foundation Model Integrating Visual Understanding and Generation [ICLR 25]
- World Model on Million-Length Video and Language with Blockwise RingAttention [ICLR 25]

Continue Tokenizer (e.g., Diffusion Loss)

- Autoregressive Video Generation without Vector Quantization [ICLR 25]

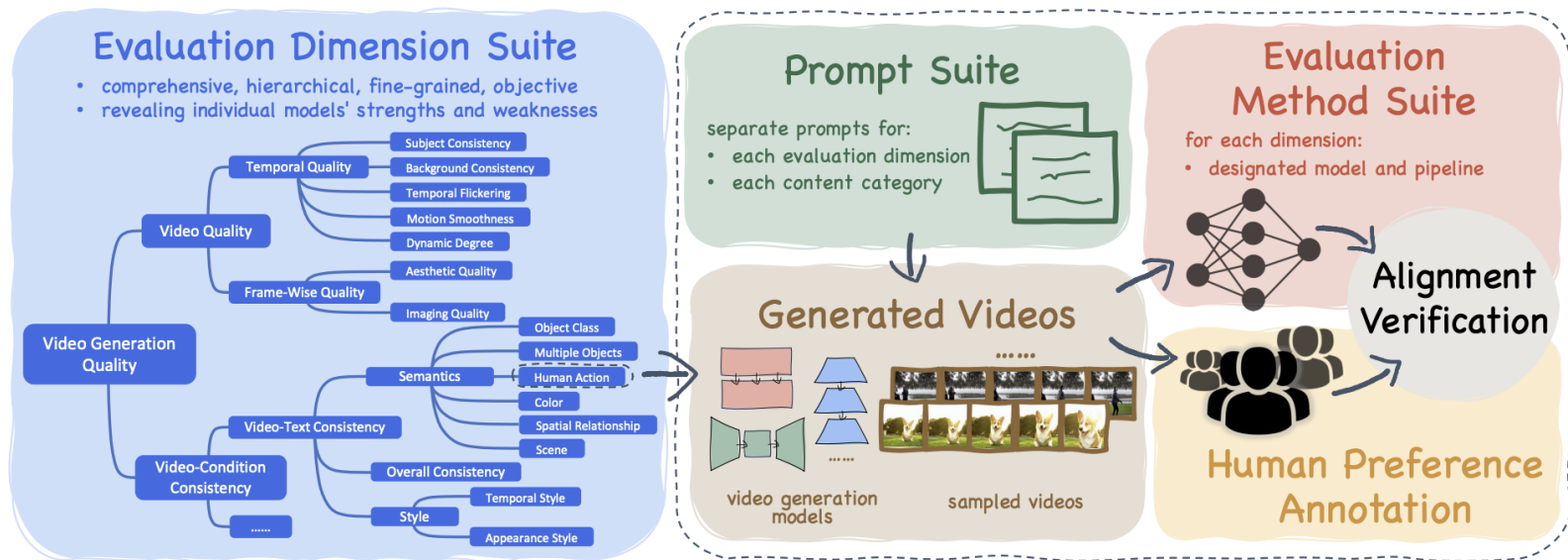
Token condition + Diffusion decoder

- NEXT-GPT: Any-to-Any Multimodal LLM [ICML 24]
- WeGen: A Unified Model for Interactive Multimodal Generation as We Chat [CVPR 25]

* Generation

+ Unified Models' Video Generation Benchmarks

× *Vbench*

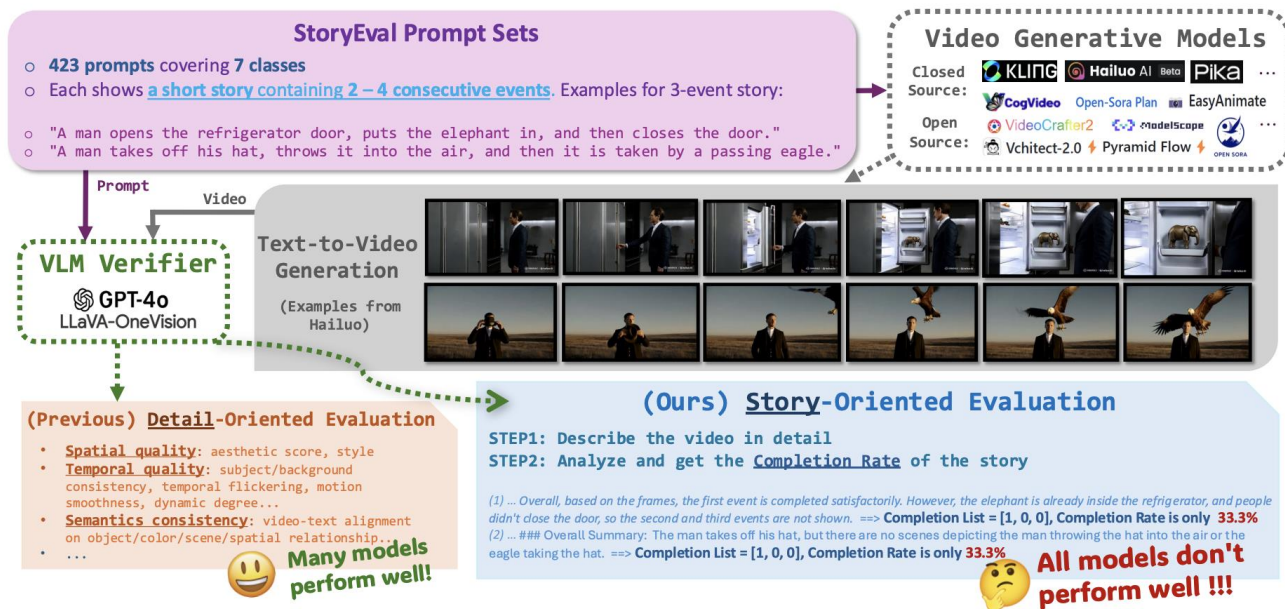


Advantages: fine-grained evaluation, multi-dimensional combination, combining automatic and manual evaluation.

* Generation

+ Unified Models' Video Generation Benchmarks

× StoryEval

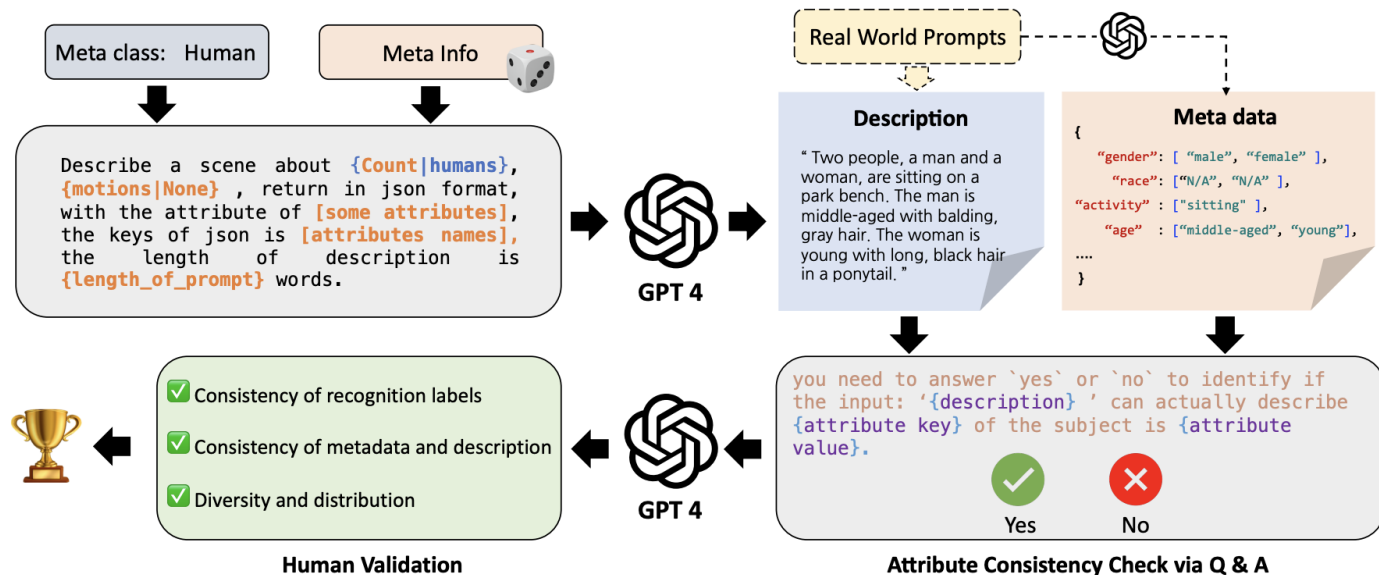


Advantages: Evaluate the “storytelling/world simulation” capabilities of video generation, focusing on event coherence and semantic consistency. Applicable to multi-event, complex scene generation evaluation.

* Generation

+ Unified Models' Video Generation Benchmarks

× *EvalCrafter*

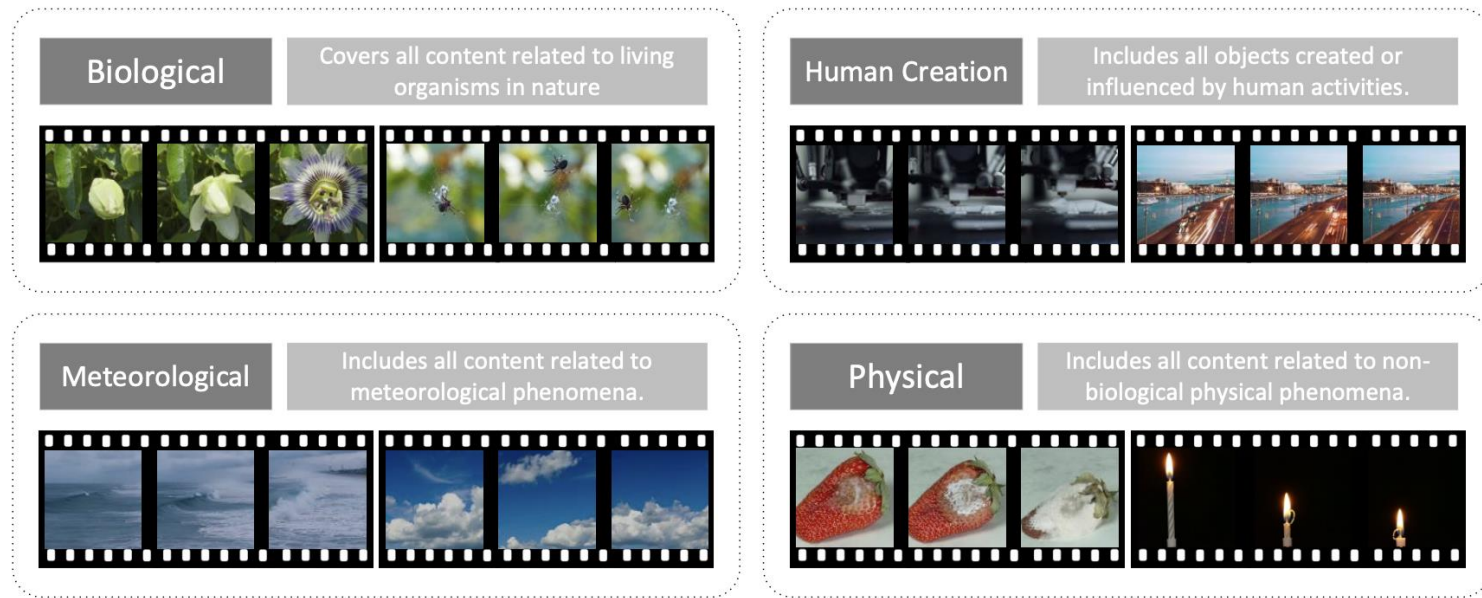


Advantages: Focus on core model capabilities such as semantic understanding, cross-modal alignment, and controllability.

* Generation

+ Unified Models' Video Generation Benchmarks

× *ChronoMaaiic-Bench*



Advantages: Focusing on the performance of video generation models in terms of temporal changes, physical processes, and temporal coherence, it is particularly suitable for testing the model's ability to capture changes in the real physical world.

* Multimodal Benchmarks of VideoLLM

+ Capability

- × Understanding
- × Reasoning
- × Generation
- × **Application**

* Application

Embodied AI Application

Third-Person View Movement

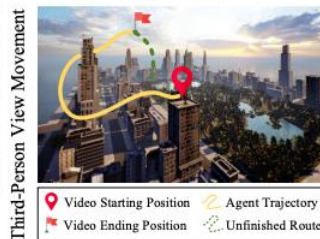
Real City



EmbodiedCity Simulator



AerialVLN Simulator



Video



[Perception] Goal Detection

Question: Navigation Goal: [Main entrance, where there is a stone]. At current position, is the destination in the field of view? If yes, where is the destination in the view?

Choices:

- A. Yes, bottom left of the field of view.
- B. Yes, centre of the field of view.
- C. Yes, top right corner of the field of view.
- D. Not in.
- E. Yes, bottom right of the field of view.

[Reasoning] Association

Question: Navigation Goal: [Main entrance, where there is a stone]. What is the most spatially related to the goal location, and where?

Choices:

- A. Periphery of the residential area, top area of the view.
- B. A green area, in the bottom right corner of the view.
- C. High-rise building, in the middle of the view.
- D. A section of road, in the right part of the view.
- E. No relative object or element.

[Navigation] High-Level Planning

Question: Navigation Goal: [Main entrance, where there is a stone]. To reach the destination from current position, what should the UAV approach next?

Choices:

- A. The ground level of the residential area
- B. The periphery of the residential area
- C. The green area
- D. The street
- E. The top of the tallest building

Agent-based Application



T.	Question	Examples
Caption	Detailed Description	Q: Please provide a detailed description of what occurs throughout these sequential GUI images. A: The video shows a user taking the 16 Personalities test on a Windows desktop using the Edge browser...
	Summarized Caption	Q: Write a clear description of the video, make sure the key features are well covered. A: Creating a new IT team in Todoist by selecting industry, job function, role, team size, and inviting members.
Static	Layout, Icon Retrieval	Q: What related searches are suggested on the right side of the Bing results for 'emnlp 2024'? A: The suggested related searches shown include 'emnlp 2024 miami', 'eac1 2024 call for papers'...
	Textual Retrieval	Q: What is the estimated time to complete the content for Week 2 of the course? A: The estimated time to complete the content for Week 2 of the course is 1 hour...
	Interrelations in GUI Content	Q: What is the name of the browser and the tab where the user performs the product search? A: The browser is Microsoft Edge, and the user performs the product search in the eBay tab.
Dynamic	Content Retrieval	Q: What specific action does the user take after turning their head to the left to view the left side of the page? A: After turning their head to the left to view the left side of the page, the user performs...
	Prediction	Q: Given the mouse is over 'Add NeurIPS 2024 DB Track Submission,' what's the likely next step? A: It would be to click on the 'Add NeurIPS 2024 Datasets and Benchmarks Track Submission' button...
	Sequential Reasoning	Q: Scrolls down from the 'Moon Gravity', which of the following cheats? A. Change Weather B. Skyfall ... A: [[B]]

UrbanVideo-Bench, EmbodiedBench ...

GUI-World, OSWORLD ...

* Summary

Four capability pillars: Understanding, Reasoning, Generation, Application form the evaluation framework.

- **Understanding** – visual perception (low-, fine-, coarse-grained) and contextual comprehension (long-context & interleaved video-language).
- **Reasoning** – relational, multi-step (CoT) & knowledge-based inference; objective vs. subjective evaluation emerging.
- **Generation** – unified models judged by VBench, StoryEval, EvalCrafter & ChronoMagic-Bench for fidelity, coherence, controllability.
- **Application** – agent-based (GUI-World, OSWORLD) and embodied-AI (UrbanVideo-Bench, EmbodiedBench) tests close-loop task execution.

Thanks!

Any questions?

